STARTER MOTOR (Continued)

INSTALLATION

2.4L

(1) Connect solenoid terminal wire harness connector to starter solenoid. Always support starter motor during this process. Do not let starter motor hang from wire harness.

(2) Install battery cable eyelet onto starter solenoid stud. Refer to Torque Specifications.

(3) Position starter motor to transmission. Install and tighten 2 bolts. Refer to Torque Specifications.

(4) Lower vehicle.

(5) Connect negative battery cable.

4.0L

(1) Connect solenoid terminal wire harness connector to starter solenoid. Always support starter motor during this process. Do not let starter motor hang from wire harness.

(2) Install battery cable eyelet onto starter solenoid stud. Refer to Torque Specifications.

(3) Position starter motor to transmission housing. Loosely install two mounting bolts.

(4) Tighten upper (rearward facing) mounting bolt to 47.5 N·m (35 ft. lbs.).

(5) Tighten lower (forward facing) mounting bolt to 40.7 N·m (30 ft. lbs.).

(6) Lower vehicle.

(7) Connect negative battery cable.

STARTER MOTOR RELAY

DESCRIPTION

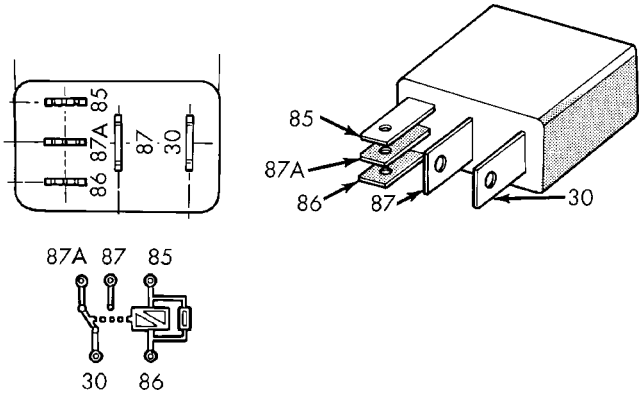

Fig. 13 Starter Relay

30 - COMMON FEED
85 - COIL GROUND
86 - COIL BATTERY
87 - NORMALLY OPEN
87A - NORMALLY CLOSED

The starter relay (Fig. 13) is an electromechanical device that switches battery current to the pull-in coil of the starter solenoid when the ignition switch is turned to the Start position. The starter relay is located in the Power Distribution Center (PDC), in the engine compartment. See the fuse and relay layout label affixed to the inside surface of the PDC cover for starter relay identification and location.

The starter relay is a International Standards Organization (ISO) micro-relay. Relays conforming to the ISO specifications have common physical dimensions, current capacities, terminal patterns, and terminal functions. The ISO micro-relay terminal functions are the same as a conventional ISO relay. However, the ISO micro-relay terminal pattern (or footprint) is different, the current capacity is lower, and the physical dimensions are smaller than those of the conventional ISO relay.

The starter relay cannot be repaired or adjusted and, if faulty or damaged, it must be replaced.

OPERATION

The ISO relay consists of an electromagnetic coil, a resistor or diode, and three (two fixed and one movable) electrical contacts. The movable (common feed) relay contact is held against one of the fixed contacts (normally closed) by spring pressure. When the electromagnetic coil is energized, it draws the movable contact away from the normally closed fixed contact, and holds it against the other (normally open) fixed contact.

When the electromagnetic coil is de-energized, spring pressure returns the movable contact to the normally closed position. The resistor or diode is connected in parallel with the electromagnetic coil in the relay, and helps to dissipate voltage spikes that are produced when the coil is de-energized.

DIAGNOSIS AND TESTING - STARTER RELAY

The starter relay (Fig. 14) is located in the Power Distribution Center (PDC), in engine compartment. Refer to fuse and relay layout label affixed to underside of PDC cover starter relay identification and location.

RELAY TEST

(1) Remove starter relay from PDC. Refer to **Starter Relay**.

(2) A relay in de-energized position should have continuity between terminals 87A and 30, and no continuity between terminals 87 and 30. If OK, go to Step 3. If not OK, replace faulty relay.

(3) Resistance between terminals 85 and 86 (electromagnet) should be 75 ± 5 ohms. If OK, go to Step 4. If not OK, replace faulty relay.

(4) Connect a battery to terminals 85 and 86. There should now be continuity between terminals 30 and 87, and no continuity between terminals 87A and 30. If OK, perform Relay Circuit Test that follows. If not OK, replace faulty relay.